

INFO/COMM 3450/INFO 5355

Group Assignment [A5]

Enrolled in [3450]

Project topic: [Problems in Living Design]

Word count: [1979]

### **Problem Statement**

An undergraduate college student who values functionality and comfort in their living space but is limited by a busy schedule needs an easy, flexible way to personalize their room while maintaining a balance between aesthetics and practicality because they want to create a relaxing environment that supports their well-being and productivity, despite time constraints and limited space.

## ***1.Report***

### **Usability Test Plan for "Smart Modular Living System"**

#### **Evaluation Goals**

1. Assess ease of use for the app's **AR visualization** feature and modular furniture assembly.
2. Evaluate the app's **AI-driven personalization** for design suggestions.
3. Identify usability challenges in the app's **community engagement** feature.
4. Measure overall user satisfaction with the system's ability to balance functionality and comfort.

#### **Step One: Plan Usability Testing Session**

##### **Metrics**

- **Quantitative:**
  - Task completion time (e.g., assembling furniture or configuring designs in the app).
  - Number of errors or instances of confusion during tasks.
- **Qualitative:**
  - Participant feedback on ease of use and frustration points (via “think-aloud” method).
  - Emotional reactions (observed body language, verbal expressions).

##### **Participants**

- **Criteria:**
  - Undergraduate students living in dorms or apartments.
  - Ages 18–22, with a mix of design experience levels (novices and experienced users).
  - Interest in sustainability or functionality in design.
- **Recruitment Message:**

*"Are you a college student looking to enhance your living space? Participate in a usability test for an innovative design solution that combines modular furniture and a digital app. Help us improve our system while learning creative design ideas. Contact us at [[edv23@cornell.edu](mailto:edv23@cornell.edu), [poa22@cornell.edu](mailto:poa22@cornell.edu), [yrv7@cornell.edu](mailto:yrv7@cornell.edu), [kfb36@cornell.edu](mailto:kfb36@cornell.edu)]. (any of them)"*

## **Time & Location**

- **Location:** Usability lab on campus or participant's dorm room or apartment for "in-the-wild" testing. (better if the place is the participant's living space).
- **Duration:**  $\leq 20$  minutes per session.

## **Data Collection Techniques**

- Video/audio recording of interactions.
- Note-taking for observed behaviors and comments.
- Screen capture for app interactions (or video by mounting a camera to capture the whole interaction - don't film the participant, but film their interaction).

## **Materials**

- Figma Prototype.
- Recording equipment, notebooks, pens.
- Task instructions for participants and observers.

## **Step Two: Develop Usability Testing Script**

### **Tasks - Participants**

#### **1. Use AR Visualization (AR Screen):**

Task 1: Scan your room using the app and find the necessary instructions to implement the "Studios" layout.

Task 2: Choose a photo from your camera roll, and find the necessary instructions to implement the "Spacious" layout.

#### **2. Personalize Design with AI (Chat AI):**

Task 3: Find the "beach vibes" prompt with the app's AI suggestions.

#### **3. Engage with the Community Feature (Feed):**

Task 4: View room setups shared by others, and find the direct message from Kendrick.

## **Tasks - Evaluators/Observers**

**Task 1:** Scan your room using the app and find the necessary instructions to implement the “Studious” layout.

**Starting point:** Scan the room by tapping the “take photo” button at the bottom of the screen on the homepage.

**End point:** Navigate to the Instructions screen.

**Path:** Take Photo → Choose desired design → Implement. The user has to see the following instructions:

- Identify areas where space is limited or feels cluttered to determine the best location for the chair.
- Select a corner, wall, or underused area where the chair can be moved without obstructing pathways or other furniture.
- Adjust nearby furniture pieces, like tables or shelves, to maximize the open space created by moving the chair.
- If needed, tweak the arrangement slightly to achieve the best balance of space-saving and practicality.

**Instructions:** The user has to start their interaction on the home screen of the app.

**Task 2:** Choose a photo from your camera roll and find the necessary instructions to implement the “Spacious” layout.

**Starting point:** Tap the Camera Roll button on the bottom left corner of the homepage.

**End point:** Navigate to the Instructions screen.

**Path:** Tap Camera Roll button → Choose a photo of the room to implement → Choose the desired design to implement → Implement. The user has to see the following instructions:

- Decide on the best position for the lamp near the desk where it can provide adequate lighting for your work or study area.
- Locate a nearby electrical outlet near the desk to plug the lamp in once it’s moved.
- Organize any nearby items around the desk to ensure the space remains functional and visually appealing.

**Instructions:** The user has to start their interaction on the home screen of the app.

**Task 3:** Find the “beach vibes” prompt with the app’s AI suggestions.

**Starting point:** Tap the AI Chat button on the top left corner of the homepage.

**End point:** Navigate to the Beach Vibe Prompt.

**Path:** Tap the AI Chat button → Tap the menu button on the top left corner → Select the Beach

Vibe Prompt. The user has completed the task when they see their beach vibe prompt starting with the text: “Help me create a beach vibe in my room.”

**Instructions:** The user has to start their interaction on the home screen of the app.

**Task 4:** View room setups shared by others and find the direct message from Kendrick.

**Starting point:** Tap the Feed icon on the top right corner of the homepage.

**End point:** Navigate to the Messages screen with Kendrick.

**Path:** Tap the Feed icon on the top right corner → Tap the Messages icon on the top right corner → Select Kendrick’s message. The user has completed the task when they see their last conversation with Kendrick: “Cool setup but maybe try the dresser in different positions.”

**Instructions:** The user has to start their interaction on the home screen of the app.

## Questions

- **Pre-task:**
  - "How do you currently design or organize your living space?"
  - "Are you familiar with apps or tools for interior design?"
- **Post-task:**
  - "What did you find easy in these tasks? What did you find challenging?"
  - "What improvements would you suggest for the app?"
  - “What tools could help you achieve the task’s goals easier?”
  - “With what tasks did you struggle with the most? Why?”
  - “Did you find the prototype to match familiar design patterns?” Is the design of the app something you’ve seen before on another app?”
  - “Did you struggle to understand the homepage?”

## Session Protocol

1. Welcome and introduce the session purpose.
2. Provide informed consent and explain recording.
3. Guide participants through tasks, encourage the “think-aloud” method.
4. Conclude with debriefing questions and thank participants.

**Note:** In composing the session protocol, pay attention to ethical issues, specifically, making the participant feel comfortable that it is not them being evaluated, but the design; any mistakes they make in using the prototype is a result of poor design and not because they are incompetent. Follow standard human participants protocols covered in the user research stage of the human-centered design process.

## Step Three: Run Usability Testing Sessions

## Lab Setup

- Observer sits diagonally behind participant; facilitator beside participant.

## Roles

- **Facilitator:** Introduces tasks and manages session flow.
- **Note-taker/Observer:** Records user behaviors, comments, and emotional responses.

## Pilot Session

- conduct pilot session before the actual usability test to make sure that everything is in order and there is no need for corrections;

## Step Four: Analyze the Data (After usability test)

1. Transcribe recordings, annotate key points.
2. Categorize data into usability issues (e.g., navigation confusion, task difficulty).
3. Summarize metrics:
  - Average task completion time.
  - Frequency of errors or frustrations.
  - Themes from qualitative feedback.
4. Identify high-priority design problems and propose solutions.

## Justification of Plan and details

The usability test plan for the "Room Vision" evaluates key features like AR visualization, AI personalization, and community engagement to meet the persona's needs for functionality, comfort, and sustainability. It focuses on ease of use, identifying challenges, and measuring satisfaction to refine the design.

Metrics combine quantitative measures (task completion time, error count) and qualitative insights (think-aloud feedback, emotional responses) for a comprehensive understanding of usability. Participants—college students aged 18–22—are selected to match the persona's demographics, with testing conducted in real-world settings for authenticity. A Figma prototype enables cost-effective, realistic interactions.

Tasks reflect real scenarios, such as arranging furniture or exploring AI suggestions, with pre- and post-task questions uncovering usability challenges. The usability testing protocol ensures a structured and thorough evaluation, focusing on capturing both user interactions and feedback effectively. A structured setup, defined roles, and a pilot session ensure smooth execution. Data collection methods like recordings and notes allow detailed analysis to identify design problems and improve the app. This approach ensures the prototype evolves to effectively support its users.

## Justification of three most significant changes

### 1.Implementation of a new Home Page:

- **Before:** The original home screen, inspired by Snapchat's camera interface, offered four paths (AI Chat, Feed, Scan Room, Upload Picture) but caused confusion during usability tests. Users struggled to understand the design, which violated **Consistency and Standards** by not following platform conventions, **Error Prevention** by lacking clear guidance, and **Help and Documentation** by not providing usage instructions.
- **After:** The redesigned home screen resolves these issues with a more intuitive layout, including a navigation bar visible across all screens, a prominent "Scan Room" button, and an "Upload Picture" button. Clear structure and popup instructions provide step-by-step guidance, aligning with standard design patterns and preventing errors. This transformative update simplifies navigation, enhances clarity, and empowers users to engage with the app confidently and efficiently.

### 2.Changed icons:

- **Before:** The Feed, Add User, and Messages icons on the Feed screen (with the Feed icon visible throughout the prototype) did not adhere to common design practices or platform conventions. As a result, users in our usability tests struggled to understand the prototype and its functionality. They expressed confusion about what specific icons represented or the actions they triggered. This lack of clarity negatively impacted the user experience, leaving users unsure about how to navigate the app properly. The violated heuristic was **Consistency and Standards**, as users had to guess the purpose of each icon and the actions it represented.
- **After:** We replaced the Feed, Add User, and Messages icons with more intuitive and representative alternatives that clearly communicate their purpose. While the flow of interaction remains unchanged, the updated icons adhere to common platform conventions, allowing users to immediately recognize their functionality. During the redesign, we carefully considered trade-offs and alternative solutions to select icons that best suit user interactions and enhance usability. This improvement prevents errors and enhances the overall user experience, making navigation more intuitive. Familiar iconography helps users quickly understand how the app works by aligning with their expectations of standard platform conventions, enabling seamless interaction with the interface

### 3.Developed Interactivity:

- **Before:** The prototype lacked fully developed features and a clear structure, leading users to encounter “Feature is Under Development” screens during tasks. This caused confusion and prevented them from completing the “happy path.” The violated heuristics were **Consistency and Standards** and **User Control and Freedom**, as participants struggled to navigate and complete their tasks.
- **After:** We fully implemented the AI Chat, AR, and Feed screens, ensuring users could experience complete interactions. Enhancements include a customized prompt with a “friend” on the Feed screen and a personalized task on the AI Chat screen that progresses dynamically. These updates align with platform conventions, improve consistency, and enable users to easily follow the “happy path,” significantly enhancing task completion and overall usability.

### Team Members contributions

1. **Eric:** Implemented the full extent of the AR feature of the app and reconstructed the design of the prototype. Worked on the report and evaluated its cohesiveness.
2. **Pedro:** Implemented the full extent of the Feed feature of the app and made it fully responsive.
3. **Kai :** Implemented the full extent of the AI chat feature of the app by adding a sample prompt for interaction.
4. **VJ:** Worked on the report, made sure everything is ready for submission, and filmed the video.

2. *Merged Heuristic Evaluation Spreadsheet:* [Merged Heuristic Evaluation Spreadsheet](#)

3. *Merged Usability Test Analysis Spreadsheet:* [Merged Usability Test Analysis Spreadsheet](#)

4. *Figma prototype:* [Before-Mid Fidelity Prototype](#)  
[After-High Fidelity Prototype](#)

5. *Showcase movie:* [Showcase movie](#)